

MARANDA 2025 KCSE PREDICTION



EXPECTED EXAM 1



PHYSICS

232/3

PAPER 3 (PRACTICAL)

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES:-

- Write your **name**, **index number** and **school** in the spaces provided above.
- Answer **all** the questions in the spaces provided in the question paper
- You are supposed to spend the first **15 minutes** of 2 ½ hours reading the whole paper carefully before commencing your work.
- Marks are given for a clear record of the observations actually made, their suitability, accuracy and the use of them
- Record your observations as soon as you make them.
- Mathematical tables, slide rules and silent non-programmable electronic calculators may be used.

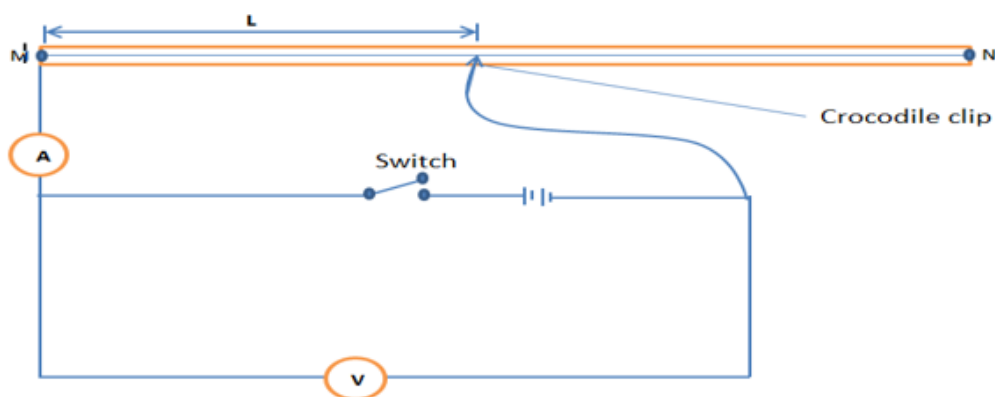
Take $g = 10\text{ms}^{-2}$

FOR EXAMINERS' USE ONLY:

QUESTION	MAXIMUM SCORE	CANDIDATE SCORE
1	20	
2	20	
TOTAL SCORE	40	

QUESTION 1

- a) You are provided with the following apparatus:
- Resistance wire fitted on a millimeter scale labeled MN
 - Switch
 - Voltmeter (0-3v) or(0-5v)
 - Ammeter (0-2.5) or(0-3)
 - Two dry cells
 - Six connecting wires
- i) Set up the apparatus as shown in the fig 1 below.



- ii) Remove the crocodile clip from the resistance wire MN and close the switch. Record the voltmeter reading.

$V_0 = \dots\dots\dots V$ (1mk)

- iii) Attach the crocodile clip to the resistance wire such that $L=0$.

Read and record the ammeter reading at $L=0$

$I_0 = \dots\dots\dots A$ (1mk)

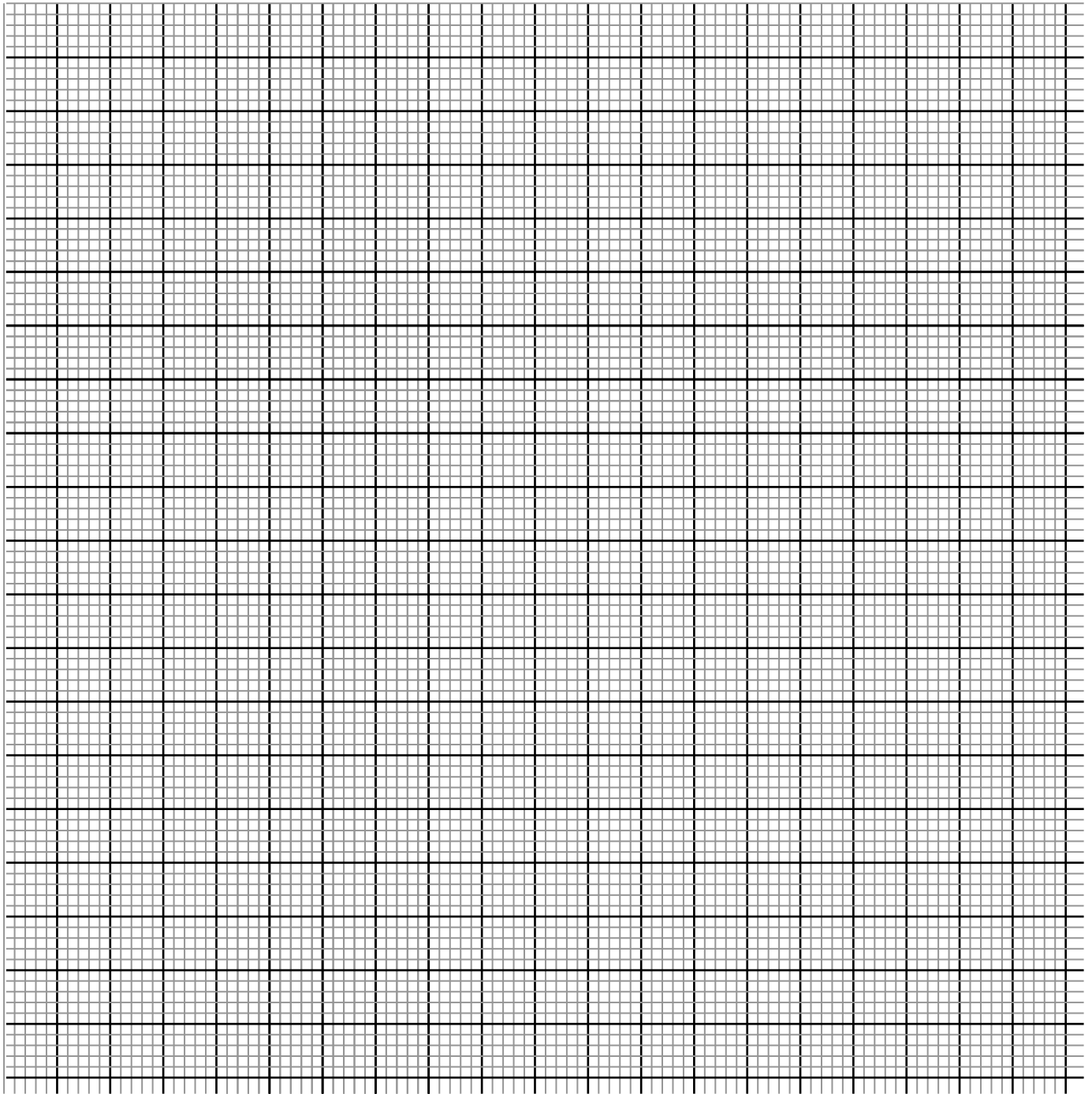
- iv) Repeat the procedure in (iii) and for $L=10\text{cm}$, 20 cm , 30 cm , 40 cm , 50cm , 60 cm , 70cm and 80 cm and record the voltmeter and ammeter reading in the table below.

- v) Complete the table below. (7marks)

Length L(cm)	10	20	30	40	50	60	70	80
Current I(A)								
p.d V(V)								
$V_0 - V$								
$\frac{V}{V_0 - V}$								
$\frac{V}{I} = R (\Omega)$								

vi) a) Plot the graph of $\frac{v}{v_0 - v}$, vertical axis against R

(5marks)



b) Determine the slope of the graph

(3marks)

c) Given that $\frac{V}{V_0 - V} = \frac{MR}{5} + d$

Where M and d are constants, Determine the values of M and d from the graph. **(3marks)**

QUESTION 2

PART A

You are provided with the following.

- A concave mirror
- A mirror holder.
- A screen
- A meter rule
- A candle
- A match box (to be shared)

Proceed as follows:

- a. i) Using the concave mirror provided focus a sharp image of a distant object (a laboratory window) on the screen. Measure the distance (X) between the mirror and the screen at which a sharp image is obtained repeat this three times, and record your readings in the table below 3marks

TRIAL	Distance(X) (cm)
1	
2	
3	

- ii) Calculate the average value of X. **(2 marks)**

- iii) What is the physical significance of the result obtained in (ii) above? **(1 mark)**

b) Set up the apparatus as in figure 2 below.

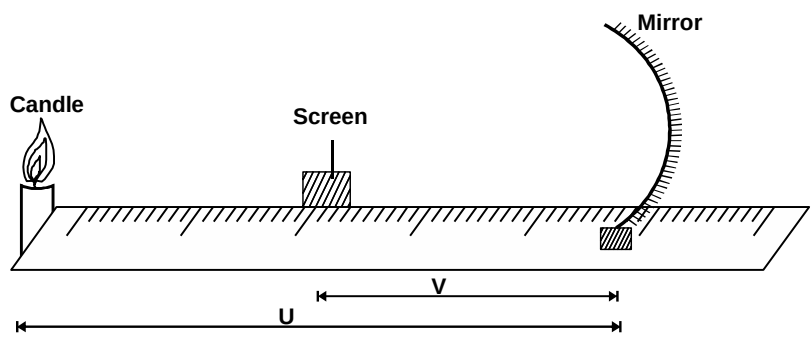


Fig 2

- c) Put the object at a distance $u = 30\text{cm}$ and from the concave mirror. Adjust the position of the screen until a sharp image is formed on the screen. Record the distance V
- d) Repeat procedure (c) above for the distance $u = 40\text{cm}$ and record the new distance V , in the table below
- e) complete the table below.

3marks

U(cm)	V (cm)	$m = V/U$	$(m + 1)$
30			
40			

f) Given that _____, calculate the values of f in each case; (3 marks)

2 PART B

You are provided with the following :

- a boiling tube
- a measuring cylinder
- a half metre rule
- water in a container
- a stand complete with boss and clamp
- vernier callipers (may be shared)

Proceed as follows:

- a) i)** Using the vernier callipers measure the internal diameter,(d),of the boiling tube provided

d =..... cm (1/2 mark)

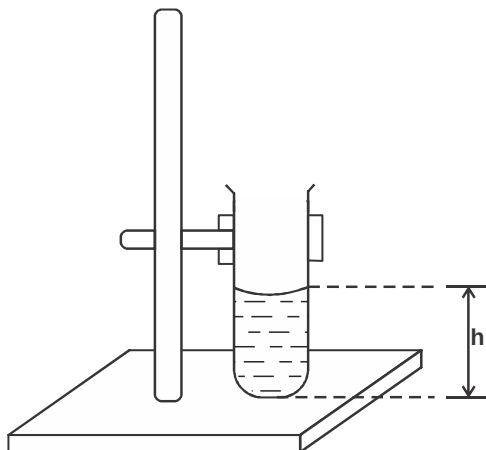
- ii)** Using the half meter rule measure the height of the boiling tube provided

h =.....cm (1/2 mark)

- iii)** Calculate the volume of boiling tube

(2marks)

- b) i)** Clamp the boiling tube vertically as shown in the figure 1 below.



ii. Using the measuring cylinder pour 25cm^3 of water into the boiling tube. Measure and record in the table below the height h , of water in the boiling tube.

iii) Repeat the procedure in b (ii) for other volumes of water, V , shown in the table. **3marks**

Volume, V , of water (cm^3)	Height, h , of water (cm)	$S = V/h$
25		
35		
45		

iv) Calculate the average value of S and state what it represents
2marks